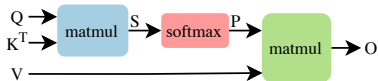


$$\text{softmax}(QK^T)V$$



```
for all  $i, j \in [0, N) \times [0, N)$  do
  for  $k \in [0, d)$  do
     $S_{i,j} := S_{i,j} + Q_{i,k}K_{j,k}$ 
```

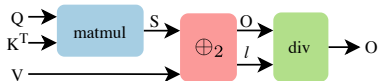
```
for all  $i \in [0, N)$  do
  for  $j \in [0, N)$  do
     $\tilde{m}_i := \max(m_i, S_{i,j})$ 
```

```
for all  $i \in [0, N)$  do
  for  $j \in [0, N)$  do
     $P_{i,j} := \exp(S_{i,j} - \tilde{m}_i)$ 
     $\ell_i := \ell_i + P_{i,j}$ 
     $P_{i,j} := P_{i,j} / \ell_i$ 
```

```
for all  $i, j \in [0, N) \times [0, d)$  do
  for  $k \in [0, N)$  do
     $O_{i,j} := O_{i,j} + P_{i,k}V_{k,j}$ 
```

dependent reductions make  
j-loop unfusable

$$\text{fused\_softmax\_matmul}(QK^T, V)$$



```
for all  $i, j \in [0, N) \times [0, N)$  do
  for  $k \in [0, d)$  do
     $S_{i,j} := S_{i,j} + Q_{i,k}K_{j,k}$ 
```

```
for all  $i \in [0, N)$  do
  for  $j \in [0, N)$  do
```

```
     $\tilde{P} = e^{S_{i,j}}$ 
     $\begin{bmatrix} m_i \\ \ell_i \\ O_i \end{bmatrix} := \begin{bmatrix} m_i \\ \ell_i \\ O_i \end{bmatrix} \oplus_2 \begin{bmatrix} S_{i,j} \\ \tilde{P} \\ \tilde{P}V_j \end{bmatrix}$ 
```

```
for all  $i, j \in [0, N) \times [0, N)$  do
   $O_{i,j} := O_{i,j} / \ell_i$ 
```

fuse  $S_{i,j}$  producer into  $\oplus_2$   
fuse  $O_{i,j}$  consumer into  $\oplus_2$   
tile by  $(B_r, B_c)$   
decomposeOp

```
for all  $i \in [0, \lfloor \frac{N}{B_r} \rfloor)$  do
  for  $j \in [0, \lfloor \frac{N}{B_c} \rfloor)$  do
    for all  $i_t \in [0, B_r)$  do
      for  $j_t \in [0, B_c)$  do
```



```
for all  $i, j \in [0, N) \times [0, N)$  do
  for  $k \in [0, d)$  do
     $S_{i,j} := S_{i,j} + Q_{i,k}K_{j,k}$ 
```

```
for all  $i \in [0, N)$  do
  for  $j \in [0, N)$  do
```

```
     $\tilde{m} = \max(m_i, S_{i,j})$ 
     $\tilde{P} = e^{S_{i,j} - \tilde{m}}$ 
     $\tilde{\alpha} = e^{m_i - \tilde{m}}$ 
     $\tilde{m}_i := \tilde{m}$ 
     $\ell_i := \alpha \ell_i + \tilde{P}$ 
```

```
    for all  $k \in [0, d)$  do
       $O_{i,k} := \alpha O_{i,k} + \tilde{P}V_{j,k}$ 
```

```
for all  $i, j \in [0, N) \times [0, d)$  do
   $O_{i,j} := O_{i,j} / \ell_i$ 
```

≈

FlashAttention-2